

1 Introduction

The goal of the project is to use the techniques learned during the course to study a problem in nonlinear dynamics both analytically and numerically. There are only a limited number of different projects, so multiple students will do the same project. You are allowed to talk to each other about the project and help each other, but each write up should reflect the work of that one individual, and the code, analysis and figures should be written by that person, and will be graded separately. The difficulty of the projects vary, choose the one that matches your ability best. The deadline for handing in the project (write-up including figures as pdf file and in hard copy; your codes in a zip file) is **23:59, June 19th, 2026**. Note: I want to grade anonymously, please only put only your student number (not your name) in the file names and the title page of your write-up.

Each project (write up) consists of the following parts

- a-10pts Look up and read the primary literature on the problem and find out what it has been used for. This is your introduction. [<200 words]
- b-10pts Determine what are the appropriate numerical methods to solve the map and/or differential equations and implement them in your preferred language (i.e. matlab or python) and test the methods for accuracy. This will be part of your methods section. [<200 words, + separate code]
- c-20pts Perform analytical analysis, such as, determine nullclines, fixed points and stability thereof. [<1000 words, <3 figures]
- d-10pts Describe and implement the methods needed for analysis of your numerical results (bifurcation diagrams, dimension, strange attractor) [<300 words, +code]
- e-40pts Perform numerical analysis, confirm your analytical results, find the strange attractors etc. as appropriate [<1000 words, <8 figures]
- f-10pts Discussion and conclusion. Summarize your results and put them in perspective of the primary literature. Comment on what you have learned. [<300 words]

For some of the more advanced projects (7,11,12) you may need much more text and figures to properly describe the results under (c) or (e), contact me if this is the case, so we can come to a solution.

1.1 Project 1: The Logistic map

The logistic map, given by

$$x_{n+1} = rx_n(1 - x_n) \tag{1}$$

looks deceptively simple. As was discussed during the lecture it has simple behavior for $r \leq 1$, as it converges to zero, but for larger values a nonzero

fixed point appear, followed by a sequence of period-doubling bifurcations with bifurcation values r_n (for the n th period doubling) that converge towards $r_\infty = 3.5699456718$ in a geometric fashion $(r_n - r_{n-1})/(r_{n+1} - r_n) \rightarrow 4.669201$. The key primary literature on this topic is old: the May et al Nature (1976), p.45-67 and the Feigenbaum (1978/1979) papers. The latter two papers are quite technical and is not recommended, the former is quite readable. Your analytical goals are:

- 1 to find behavior and stability of the fixed point before the onset of period doubling.
- 2 to determine to onset of period doubling by determining where the single fixed point loses stability

Your numerical goals are

- 1 Implement map equations and test the correctness by comparing to fixed point calculated analytically
- 2 Construct the bifurcation diagram by iterating the equation for each different r value
- 3 Determine Lyapunov exponent as a function of r [Consult Strogatz for this calculation]
- 4 Demonstrate the self-similarity of the bifurcation diagram
- 5 Study the attractor that emerges for $r = 4$. Demonstrate that the dynamics is chaotic and show that the dimension of the attractor is about one (note that convergence can be very slow...)

1.2 Project 2: The Henon-Heiles map

The Henon-Heiles map is the most studied example of a two-dimensional chaotic map. The equations are given by

$$X_{n+1} = 1 - aX_n^2 + bY_n \quad (2)$$

$$Y_{n+1} = X_n \quad (3)$$

a with standard parameters $a = 1.4$ and $b = 0.3$.

- 1 Determine for which values of a and b there are stable fixed points.

Your numerical goals are

- 1 Implement the map equations and test the correctness by comparing to your fixed points calculated analytically for a few a and b combinations.
- 2 Pick initial conditions X_0 and Y_0 that do not run off to infinity and run the iterations in the standard case. Show that the iterates remain bounded, but that they *look* chaotic.

- 3 Show that the solutions are chaotic by demonstrating sensitivity to initial conditions.
- 4 Determine the resulting strange attractor, starting from X_0 and Y_0 , after discarding an appropriate iterates as transient.
- 5 The strange attractor has double lines as a structure, show that each line look itself like a double line, etc, i.e. it is a fractal structure.
- 6 Determine the basin of attraction of the attractor, i.e. those initial values that do not run off the infinity. Compare the image of the basin to the strange attractor you found in (4).

1.3 Project 3: The Lorenz attractor

The Lorenz attractor was extensively discussed during the lectures and in Chapter 9 of Strogatz. You should discuss what the motivation for the equations where, perhaps discuss some physical implementations. The book by Gleick in 1987 (Chaos: making a new science) gives a bit of the more popular science background. The equation for it is:

$$\frac{dx}{dt} = \sigma(y - x) \quad (4)$$

$$\frac{dy}{dt} = -xz + rx - y \quad (5)$$

$$\frac{dz}{dt} = xy - bz \quad (6)$$

with standard parameters $\sigma = 10$, $b = \frac{8}{3}$.

Your analytical goals are:

- 1 Determine the position of the three fixed points as a function of r and determine their stability.

Your numerical goals are:

- 1 Implement a routine to integrate the differential equations. You can use a matlab/python function or implement a Runge-Kutta algorithm from scratch.
- 2 Test your routine by reconstructing the bifurcation diagram for $r < 24.74$. This should match your analytical answers.
- 3 Extend the bifurcation diagram to $r = 130$.
- 4 For $r > 24.74$ there is chaotic dynamics. Take $r = 28$ and demonstrate chaos by showing sensitivity to initial conditions.
- 5 Determine the largest Lyapunov exponent to demonstrate chaos as well.
- 6 Characterize the strange attractor. Is it fractal?
- 7 Determine the tentmap (a sort of Poincaré map) by plotting in a return map the z values at which the trajectory crosses from one wing to another.

1.4 Project 4: The Ueda oscillator

In the past there has been a lot of interest in driven and non-driven relaxation oscillators, leading to the van der Pol oscillator, as well as the Duffing model. One popular example, which also showed chaos, is the Ueda oscillator. Discuss the prior literature on relaxation oscillators (this includes the 1930s, prior to the observation of chaos). Its equations are given:

$$\frac{d^2y}{dt^2} + b\frac{dy}{dt} + y^3 = A\sin\Omega t \quad (7)$$

with standard parameters $A = 7.5$, $b = 0.05$ and $\Omega = 1$. Your analytical goals are:

- 1 Determine the position and stability of the fixed points in the absence of external drive.
- 2 Argue whether you would expect a limit cycle to emerge (in the absence of external drive, but with b varying)

Your numerical goals are:

- 1 Implement a routine to integrate the differential equations. You can use a matlab/python function or implement a Runge-Kutta algorithm from scratch.
- 2 Check your algorithm by confirming your analytical estimates for the fixed points and their stability.
- 3 Characterize the dynamics in the absence of external drive. (I.e. follow curves in the phase plane for different initial conditions, to see what the basin of attraction for the fixed points is and find stable or unstable limit cycles if they are there).
- 4 In analogy to a pendulum, you would expect simple harmonic oscillations for small drive amplitudes. Increase the amplitude A from 0 to 7.5 and describe and interpret your observations.
- 5 Determine whether for the standard parameters there is chaos. I.e. is there sensitivity to initial conditions?
- 6 Determine the Lyapunov exponent.
- 7 Determine the Poincaré map: the value of y and dy/dt for t at multiples of the drive period. Is it fractal? (You can also use the Poincaré map in your answer to point 4)

1.5 Project 5: Bursting in Neurons

Neurons produce so called action potentials during which the membrane potential V (taken relative to the outside of the cell) switches from negative values of around -65 mV to positive values for a time interval of on the order of ms. A train (periodic sequence) of action potential represents a limit cycle. Sometimes there is no limit cycle but the neuron can still be perturbed by a short current pulse to produce a spike after which it returns to its stable equilibrium position. The neuron can be brought from a quiescent state without spiking to a spiking state by increasing a bifurcation parameter known as the driving current I . In this project you will first consider a 2-dimensional model neuron for which you will study the bifurcation from quiescent to spiking and back, after which you add a third variable representing a slower current to make the model burst, that is produce a periodic sequence of spikes, followed by quiescence, followed by spiking again, and so on.

The full 3-dimensional model has as variables V (membrane potential), n (fast recovery variable, potassium current), n_s (slow recovery variable responsible for bursting).

$$C\dot{V} = I - g_L(V - E_L) - g_{Na}m_\infty(V)(V - E_{Na}) - g_Kn(V - E_K) \quad (8)$$

$$-g_{slow}n_s(V - E_K) \quad (9)$$

$$\dot{n} = (n_\infty(V) - n)/\tau_n(V) \quad (10)$$

$$\dot{n}_s = (n_{s,\infty}(V) - n_s)/\tau_{slow}(V) \quad (11)$$

With the following parameters: the capacitance $C = 1$, conductances $g_L = 8$, $g_{Na} = 20$, $g_K = 9$, $g_{slow} = 5$, the reversal potentials are $E_L = -80$, $E_{Na} = 60$, $E_K = -90$ and $I = 5$. The dynamics of the gating variables n and n_s are expressed in terms of helper functions of the form $f_\infty = (1 + \exp((V_{half} - V)/k))^{-1}$, with parameters k and V_{half} . For n_∞ , $k = 5$, $V_{half} = -25$, whereas for m_∞ , $k = 15$, $V_{half} = -20$ and for $n_{s,\infty}$, we have $k = 5$, $V_{half} = -20$. The time scales could in principle depend on V but are here taken to be constant $\tau_n = 0.152$ and $\tau_{slow} = 20$. The membrane potential is in mV, time in ms, the capacitances are in nF, currents in nA, the conductances are in uS (You should check this).

Your analytical goals are to

- 0 Check whether the units in the equations are chosen appropriately
- 1 find the null clines for different values of n_s (which is so slow that it can be taken as constant in the phase plane analysis for the two faster variables V and n , and basically acts like I) and determine the stability of the fixed points (you can use the computer to calculate this but derive the expression analytically)
- 2 determine for which value of n_s you would expect limit cycles
- 3 characterize the type of the bifurcation from quiescent to spiking

- 4 characterize the type of bifurcation from spiking to quiescent
- 5 explain how the burst develops in the full model, when n_s is allowed to vary.

Your numerical goals are

- 1 Implement the 2-variable and 3-variable model in matlab (or python) using either your own integrator and test the accuracy against a library routine (i.e. in matlab ode45). Choose which one you want to continue to work with.
- 2 Plot trajectories of the neuron model in the phase plane for relevant values of n_s (use the analytical work to guide you) together with the null clines and the fixed points.
- 3 Determine numerically for which values of n_s (as a constant) there are spikes and calculate their rate
- 4 Determine numerically for which values of I the full model bursts, and if so, what the burst rate is and how many spikes there are in a burst. These quantities vary with I so find a way of plotting the results.
- 5 Explore – can you find chaos (it should be possible, but I did not look at it myself)? Otherwise find some other interesting property like phase locking when you apply a periodically varying current.

1.6 Project 6: Homeostasis in neural networks

Neurons spike in response to time-varying sensory inputs. On a longer time scale, they try to maintain a certain level of activity. They need to be active in order to keep synapses functional, but at a moderate level to keep metabolic demands within a reasonable range. Ensuring homeostatic stability and proper dynamical range for representing sensory inputs is difficult. Here we will look at a network model to see how that could happen. The network dynamics is represented by the activity of excitatory neurons E and that of inhibitory neurons I .

$$\tau_e \dot{E} = -E + g_e [e_e + J_{ee}E - J_{ei}I - S_e]_+ \quad (12)$$

$$\tau_i \dot{I} = -I + g_i [e_i + J_{ie}E - J_{ii}I - S_i]_+ \quad (13)$$

In this expression $[x]_+$ denotes a rectifying function $\max(0, x)$, τ_i and τ_e are the time scale of inhibitory and excitatory activity, respectively, whereas e_i and e_e are the drive representing sensory inputs, g_i and g_e the gain of the input to the neurons, J_{kl} is the strength of the input from population l to population k , S_i and S_e represent homeostatic effects. They satisfy

$$\tau_{se} \dot{S}_E = E - E_\infty \quad (14)$$

$$\tau_{si} \dot{S}_I = I - I_\infty \quad (15)$$

$$(16)$$

where E_∞ and I_∞ are the desired set points for excitatory and inhibitory activity respectively.

Your analytical goals are

- 1 Determine the null clines, and their intersection, the fixed points for a constant S_E , S_I and input e_i, e_e .
- 2 Derive the condition for which the fixed point is stable.
- 3 Assuming that the neural dynamics of E and I (time scale τ_e and τ_i) is much faster than that of the homeostatic variables (time scale τ_{se} and τ_{si}), the fixed point solution for E and I can be directly substituted in Eq.(16). This again is 2-dimensional system. Find the fixed point and determine the conditions under which it is stable.

Your numerical goals are

- 1 Implement an integration routine to solve the full four dimensional system. Choose the appropriate parameters that satisfy the stability requirement calculated in item 2 and 3 of your analytical goals. Test your algorithm against a library routine in matlab or python as appropriate. This problem can be quite hard if your time scales τ_e and τ_{se} are very different, so make the appropriate choice.
- 2 Determine how your system reacts to a sudden change in input e_e , e_i . Stability implies that the sudden increase in firing rate is transient and limited in amplitude.
- 3 The system can be made unstable by choosing (for instance) time scales that do not satisfy the constraints in 2 and 3. First make the fast subsystem unstable. Determine what happens under a change in input.
- 4 Make the slow subsystem unstable. Determine what happens under a change in input.
- 5 Instead of S_E and S_I you could also use g_e and g_i as variables to achieve homeostasis. Formulate the appropriate slow system and test it.
- 6 Try to add a limit cycle to both subsystems, study how it interacts. I have not tried this so consult me if it does not seem to work.

1.7 Project 7: Spin torque oscillators

In recent years investigators have tried to come up with new elements to build neuromorphic devices from. Neuromorphic devices mimic the architecture of neural circuits using nanoscale solid-state physics devices and can thereby perform computations in parallel rather than serially as would be the case in conventional computers. The new elements offer the promise of more energy efficient computation. The goal of this project is to look at spin torque oscillators

(STO). Conceptually they consist of a disk of three layers: a fixed layer in which a permanent magnetic field is present (direction $\mathbf{M}$), a free layer of which the magnetization vector can vary in direction $\mathbf{m}$ and a nonmagnetic layer sandwiched in between. The freely changing magnetization depends on the current flowing through the device and the direction of the fixed magnetization. It would be helpful to be able to connect multiple STOs together, but then they need to be able to synchronize to maximize power. Here you make a start with looking at the synchronization dynamics. Warning: This project is extremely challenging – it almost broke the previous student that attempted it.

The equations for a single STO are:

$$\frac{d\mathbf{m}}{dt} = -|\gamma|\mathbf{m} \times \mathbf{H}_{eff} + \alpha\mathbf{m} \times \frac{d\mathbf{m}}{dt} + |\gamma|\beta J\mathbf{m} \times (\mathbf{m} \times \mathbf{M}) \quad (17)$$

Here $\gamma = 1.9 \times 10^{11} \text{ Hz/T}$ is the gyromagnetic ratio, $\alpha = 10^{-2}$ is the Gilbert damping parameter, and $\beta = 10/3$ represents material parameter and fundamental constants. J is the current flowing from the fixed to free layer. The effective magnetic fields is composed of the external magnetic field $H_a = 0.2$ along the x -axis, and a anisotropy field $H_k = 0.05$, also along the x -axis, together with a demagnetization field $H_{dz} = \frac{4\pi}{\mu_0} M_s = 1.6$ along the z -axis, where M_s is the saturation magnetization of the free layer. Hence,

$$\mathbf{H}_{eff} = H_a \hat{e}_x + (H_k m_x \hat{e}_x - H_{dz} m_z \hat{e}_z) \quad (18)$$

According to the dynamics the magnitude of the magnetization does not change, so we can rewrite the vector equation in terms of the two angles (on the sphere):

$$\frac{d\theta}{d\tau} = U \cos \theta \cos \phi + \alpha(H_{dz} + H_k \cos^2 \phi) \sin \theta \cos \theta \quad (19)$$

$$-V \sin \phi - H_k \sin \phi \cos \phi \sin \theta \quad (20)$$

$$\frac{d\phi}{d\tau} = \frac{1}{\sin \theta} [-U \sin \phi - (H_{dz} + H_k \cos^2 \phi) \sin \theta \cos \theta \quad (21)$$

$$-V \cos \phi \cos \theta - \alpha H_k \sin \phi \cos \phi \sin \theta] \quad (22)$$

with additional definitions $\tau = \frac{\gamma t}{1+\alpha^2}$, $U = \alpha H_a + \beta(J_d + J_a)$, $V = H_a - \alpha\beta(J_d + J_a)$, with J_a being the ac and J_d being the dc current. Currents are expressed in Ampere and magnetic field is in Tesla. This gives a dimensional time scale (τ) with units of $\frac{2\pi}{189.98}$ nanoseconds.

Your analytical goals are

- 1 Give a motivation for the Landau-Lifshitz-Gilbert-Slonczewski equation (Eq.(17) and explain where the different terms in H_{eff} come from.
- 2 Supply the missing steps in reducing the differential equation from one in terms of $\mathbf{m}$ to one in terms of the angle θ and ϕ .
- 3 Perform a phase plane analysis. There are three types of solutions, a harmonic like motion in θ and ϕ , a running motion along the ϕ axis and

an heteroclinic cycle going from one fixed point to the next (in infinite time). Find the null clines and fixed points and show these three solutions in the phase plane.

You numerical goals are

- 1 Implement an integration routine for the ϕ, θ dynamics and validate this against a library routine in matlab or python.
- 2 Vary the driving current J_d in the range from 0 to 0.04 and determine the frequency of the periodic motion that the angles perform. Plot this frequency versus J_d curve and describe the salient features.
- 3 Add a periodic driving current $J_a(t) = |J_a| \cos \Omega t$ and vary both Ω (between 0.6 and 0.9 in the tau units) as well as $|J_a|$ from 0 to 2×10^{-3} while keeping J_d at 0.01. Plot, analyze and describe the phase locking behavior.
- 4 When you couple two STOs in series, they receive a common current J that depends on the direction of free magnetization in each STO as

$$J(\tau) = J_d / (C_a - C_b \Sigma_i \sin \theta_i(\tau - D) \cos \phi_i(\tau - D))$$

. The goal is to implement this equation and find conditions under which the two STOs can synchronize. Formulate and implement a routine to calculate the $\theta_1, \phi_1, \theta_2, \phi_2$ as a function of time. Make some informed choices for C_a, C_b and D .

- 5 Explore the dynamics of two coupled STO systems – can they synchronize? (Don't go overboard with this part as it can become quite challenging).

1.8 Project 8: The Epileptor

Recently Jirsa and coworkers published a paper (On the nature of seizure dynamics, Brain (2014) 137:2210-2230) wherein they tried to model the dynamics of epileptic seizures using a simple (compared to other approaches) nonlinear model. Their approach was to take a catalogus of possible ways (i.e. bifurcations) by which seizure activity could emerge and a catalogus of bifurcations by which seizure activity could end and identify using experimental data which bifurcation would best account for the data. They found that the onset most likely was via a saddle-node bifurcation, whereas the offset was most likely via a homoclinic bifurcation (saddle collides with limit cycle). These hypotheses make clear predictions about the time course with which the amplitude and period vary near the bifurcation. The goal of this project is to reproduce the epileptor dynamics given in the original paper, identify and characterize the bifurcations quantitatively, do parameter exploration to find new behaviors. This project is experimental as it has been added this year to the project line up – consult the instructor if you are stuck or the description is unclear.

The equations describe two subsystems, one fast, x_1, y_1 , one slower x_2, y_2 and one really slow z variable that is responsible for on- and offsets. In addition

there is dummy variable u (CORRECTED 24 june 2024) that accounts for feeding back filtered activity:

$$\dot{x}_1 = y_1 - f(x_1, x_2) - z + I_{rest1} \quad (23)$$

$$\dot{y}_1 = y_0 - 5 x_1^2 - y_1 \quad (24)$$

$$\dot{z} = \frac{1}{\tau_0} (4(x_1 - x_0) - z) \quad (25)$$

$$\dot{x}_2 = -y_2 + x_2 - x_2^3 + I_{rest2} + 2u - 0.3 (z - 3.5) \quad (26)$$

$$\dot{y}_2 = \frac{1}{\tau_2} (-y_2 + f_2(x_2)) \quad (27)$$

$$\dot{u} = -\gamma(u - 0.1 x_1) \quad (28)$$

with

$$f_1(x_1, x_2) = \begin{cases} x_1^3 - 3x_1^2 & \text{if } x_1 < 0 \\ (x_2 - 0.6 (z - 4)^2)x_1 & \text{if } x_1 > 0 \end{cases} \quad (29)$$

$$f_2(x_1, x_2) = \begin{cases} 0 & \text{if } x_2 < -0.25 \\ 6(x_2 + 0.25) & \text{if } x_2 > -0.25 \end{cases} \quad (30)$$

and parameter values $x_0 = -1.6$, $y_0 = 1$, $\tau_0 = 2857$, $\tau_1 = 1$, $\tau_2 = 10$, $I_{rest1} = 3.1$, $I_{rest2} = 0.45$, $\gamma = 0.01$. Starting conditions are typically: $x_1 = 0$, $y_1 = -5$, $z = 3$, $x_2 = 0$, $y_2 = 0$, $u = 0$. Your analytical goals are (these not necessarily need to be done before the numerical ones)

- 0 Read some literature about the features of seizures and summarize them so you can look for them in your simulations
- 1 The SN bifurcation plays a key role in the onset of seizure activity. Give the normal form and characterize the behavior near the bifurcation, as a function of the bifurcation parameter (z in this model).
- 2 Identify in the model in which subsystem the SN takes place and see if you can recognize its underlying normal form from the equations
- 3 The homoclinic bifurcation plays a key role in the offset of seizure activity. Give the normal form and characterize the behavior near the bifurcation, especially the logarithmic dependence of the period on the bifurcation parameter.
- 4 Identify in the model in which subsystem the homoclinic takes place and see if you can recognize its underlying normal form from the equations

Your numerical goals are

- 1 Implement and test the system of differential equations, considering z to be constant.
- 2 Implement and test the system of differential equations, considering z to be a variable.

- 3 Design views of the phase space that make the dynamics clear, so that you can properly visualize the epileptor system.
- 4 The bifurcations can be studied by looking at the x_1 versus z plane – z is considered as a slow variable, hence for bifurcation analysis we assume z to be constant. The seizure oscillations are visible as oscillations in x_1 . The scenario is as follows. For low z values there is no stable fixed point and x_1 oscillates, when z then increases, a SN bifurcation generates a stable and unstable fixed point, creating bistability. The seizure still continues, further increasing z until a homoclinic bifurcation occurs in which the unstable fixed point (from the SN) collides with the limit cycle and destroys it. The system then falls towards the fixed point and stays there while z decreases. In time z has become so small that the SN bifurcation is approached, destroying the stable fixed point and the system has to jump to the limit cycle – seizure starts. Your job is to find the appropriate z values and visually reconstruct the scenario outlined above.
- 5 Exploration: Determine how the other parameters change the picture outlined in the previous question. I.e. try to find cases where the SN or homoclinic bifurcation does not occur anymore. Do other bifurcations take over in that case? Can you find chaos? (Don't go overboard here: I have not done this myself so I can not estimate how long this would take to do it properly).

1.9 Project 9: Chaos in simple systems (preliminary)

Researchers have looked at many different forms of simple systems that give rise to chaos. In this project you will analyze your choice of systems. For a specific choice of parameters (and in some cases initial conditions) there will be chaos, hence, one can demonstrate signs of chaos such as sensitivity to initial conditions, Lyapunov exponent and strange attractors. In some cases a Poincaré map may be feasible, for instance, when there is a periodic drive. There will be less possibilities for analytical work, however you can try to find conditions (parameter values) for which there is a fixed point, or perhaps you can demonstrate a limit cycle and uses these to test your numerical algorithm for solving the differential equation (or vice versa). In the following I give a number of models taken from the literature, together with the parameters that should give chaos. Your task is to characterize the chaotic state for these parameters, vary them a little bit to see what happens with the chaotic state and vary them to a large extent to get a simpler system that you can study analytically for fixed points (or limit cycles using Poincaré-Bendixson).

Model 1 *Forced Rayleigh oscillator*

$$\frac{d^2y}{dt^2} + \left(\frac{dy}{dt} - 4\right)\frac{dy}{dt} + y = A \sin \Omega t \quad (31)$$

with standard parameters $A = 5$, $\Omega = 4$ and initial conditions $y_0 = 1$, $v_0 = 0.8$, (v refers to the velocity $\frac{dy}{dt}$).

Model 2 *Forced conservative nonlinear oscillators*

$$\frac{d^2y}{dt^2} + \frac{y}{\sqrt{|y|}} = A \sin \Omega t \quad (32)$$

with standard parameters $A = 1$, $\Omega = 2$ and initial conditions $y_0 = 0.7$, $v_0 = 0.0$.

Model 3 *Velocity forced oscillators*

$$\frac{dy}{dt} = v + A \sin \Omega t \quad (33)$$

$$\frac{dv}{dt} = -v + (1 + 0.6y)y \quad (34)$$

$$(35)$$

with standard parameters $A = 1$, $\Omega = 1$ and initial conditions $y_0 = -2$, $v_0 = 0.0$.

Model 4 *Circulant systems*

$$\frac{dx}{dt} = f(x, y, z) \quad (36)$$

$$\frac{dy}{dt} = f(y, z, x) \quad (37)$$

$$\frac{dz}{dt} = f(z, x, y) \quad (38)$$

$$(39)$$

with $f(x, y, z) = -ax + by - y^3$ with $a = 1$, $b = 4$ and $x_0 = 0.4$, $y_0 = 0$, $z_0 = 0$.

Model 5 *Coupled van der Pols*

$$\frac{d^2x}{dt^2} + b(x^2 - 1)\frac{dx}{dt} + x = k(y - \frac{dy}{dt}) \quad (40)$$

$$\frac{d^2y}{dt^2} + b(y^2 - 1)\frac{dy}{dt} + y = k(x - \frac{dx}{dt}) \quad (41)$$

with $b = 2.2$, $k = 0.7$ and initial conditions $x_0 = 0$, $\dot{x}_0 = 1$, $y_0 = 0$, $\dot{y}_0 = 1.7$.

Model 6 *Restricted three-body problem – mass 1 negligible, mass 2 and 3 identical and dynamics restricted to the plane*

$$\ddot{x}_1 = \frac{x_2 - x_1}{R_{12}^3} - \frac{x_2 + x_1}{R_{13}^3} \quad (42)$$

$$\ddot{y}_1 = \frac{y_2 - y_1}{R_{12}^3} - \frac{y_2 + y_1}{R_{13}^3} \quad (43)$$

$$\ddot{x}_2 = -\frac{2x_2}{R_{23}^3} \quad (44)$$

$$\ddot{y}_2 = -\frac{2y_2}{R_{23}^3} \quad (45)$$

with $R_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$ being the Euclidean distance with initial conditions $x_{10} = 1, y_{10} = 0, \dot{x}_{10} = 1, \dot{y}_{10} = 0.5, x_{20} = 1, y_{20} = -0.7, \dot{x}_{20} = 0.1, \dot{y}_{20} = 0.4$.

1.10 Project 10: Neuromorphic circuit of two vd Pol oscillators (Preliminary)

The idea of neuromorphic computing is to implement neuron-like dynamics using analog circuits in order (1) to save power relative to an implementations where the ordinary differential equations are integrated on a standard central processing unit (CPU) and (2) to use neural architectures that (hopefully) can more efficiently implement difficult tasks like object recognition that challenge more traditional machine learning algorithm. A recent paper coupled two oscillators, built out of so-called programmable unijunction transistor (PUT) [www.onsemi.com/pub/Collateral/2N6027-D.PDF], using a memristor, which has a history dependent resistance, here we will use primarily two settings corresponding to a high resistance state (HRS) and low resistance state (LRS), see Figure 1. Internally these oscillators (i.e. location A1 or A2) show sawtooth type dynamics with a frequency determined by the value of the capacitors and resistors, but they communicate via pulses (i.e. location G1 or G2). When the memristor is in the HRS the two oscillators do not influence each other, but when it is in the LRS they do influence each other, and synchronize, despite the fact that they want to oscillate at different frequencies. This behavior was observed in experiment and could be reproduced numerically using the van de Pol model, as given by:

$$\ddot{x}_1 - \alpha_1(1 - x_1^2)\dot{x}_1 - \beta_1 x_1 \frac{(x_1 + \gamma_1)^2}{\gamma_1^2} = m(x, t)(x_2 - x_1) \quad (46)$$

$$\ddot{x}_2 - \alpha_1(1 - x_2^2)\dot{x}_2 - \beta_2 x_2 \frac{(x_2 + \gamma_2)^2}{\gamma_2^2} = m(x, t)(x_1 - x_2) \quad (47)$$

where $m(x, t)$ is the memristor, approximated by m_0 for $t < t_{switch}$ and m_1 for $t > t_{switch}$ with typical parameters $\alpha_1 = 3.5, \alpha_2, \beta_1 = \beta_2 = 0.1, \gamma_1 = \gamma_2 = 3.0, m_0 = 0.01$ and $m_1 = 0.1$.

The overall goal of this project is to confirm the numerical equivalence of the electronic implementation and the numerical model and study the properties of the numerical model, specifically the individual oscillators analytically and the phase locking dynamics numerically (thus reproducing the experimental results). Consult the paper for additional information on the underlying electrical circuits: Ignatov et al (2016) <https://doi.org/10.1063/1.4942832>. Your analytical goals are

- (a) Study a single uncoupled oscillator – determine whether there are fixed points and whether you would be expecting limit cycles
- (b) Determine to what extent the electronic oscillator can be modeled by the van der Pol equation – that is write down the circuit equations and analyze them.

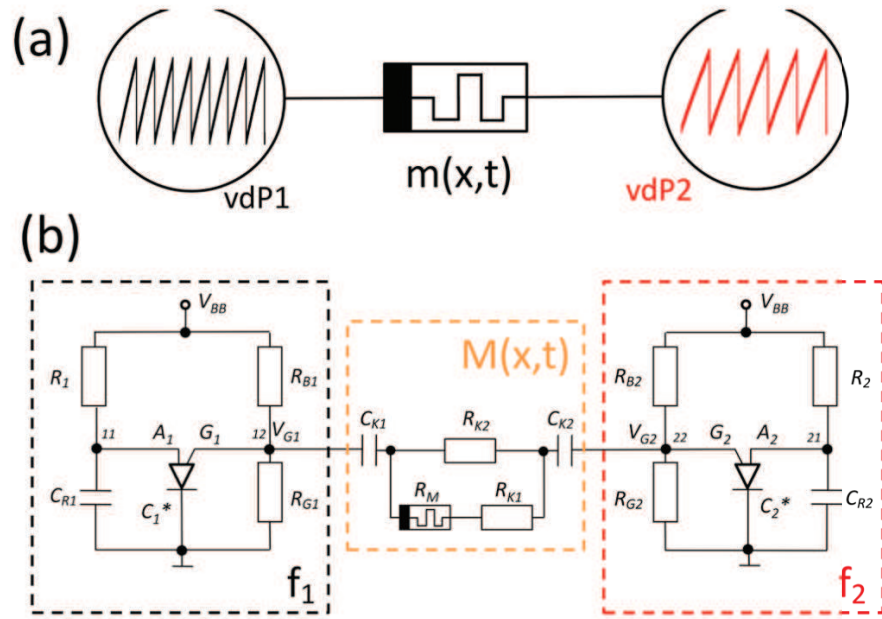


Figure 1: Neuromorphic implementation of two oscillators f_1 , f_2 coupled by a memristor M .

- (c) Look up the literature for memristors and write down some simple equations for them, described how they endow the memristor with the history-dependent properties (we will not really use them though).

Your numerical goals are

- (a) Study a single uncoupled oscillator numerically, is there a preferred frequency? Which parameters change that frequency?
- (b) Study a single uncoupled oscillator as function of the frequency and amplitude of a periodic driving force (i.e. $A \sin \omega t$, when do you observe a limit cycle?)
- (c) Determine for what connection strength the two oscillators synchronize. Make a 2D phase diagram of connection strength versus frequency difference (between the individual oscillators when uncoupled). Indicate for which values in the 2D diagram there is synchronization and for which ones there is not.
- (d) Explore parameter values to see if you can find chaotic states.

1.11 Project 11: Generalization of Kuramoto model (Preliminary)

Oscillations correspond to limit cycles. When the limit cycles are stable the dynamics can be reduced to a one dimensional variable along the limit cycle, the phase $\theta(t)$. When a perturbation is applied to the oscillator it will transiently leave the limit cycle, and when it returns to the limit cycle it will have picked up an additional phase. This means that interactions between oscillators can be described in terms of their effects on the phase. This leads to the Kuramoto equations:

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i) \quad (48)$$

In this expression ω_i is the intrinsic frequency of oscillator i , K is the strength of the coupling, which is normalized by the number of oscillators N , so that one can take the limit $N \rightarrow \infty$. The coupling is a sine of the phase difference between the pair of oscillators, which will try to make the phase difference zero (modulo 2π). Kuramoto could find an analytical solution for the degree of synchronization (full synchronization means all oscillator have the same frequency), by introducing an order parameter $r \exp(i\Theta) = (1/N) \sum_{j=1}^N \exp(i\theta_j)$, using which we get uncoupled equation for each oscillator interacting with a mean field, which then can be solved self-consistently (see the book Synchronization by Pikovsky and coworkers for this calculation). In this project we will consider to generalization of this equation. First, instead of all-to-all connectivity we will consider other networks structures such as for example a small world network. Second, a phase corresponds to a unit vector in two dimensions,

which means we can generalize these equations to the dynamics of unit vectors in higher dimensions (see Chandra and Ott, Physical Review X, 2019). Interestingly, there is a fundamental difference in synchronization properties between even and odd dimensions. This type of formulation could also be utilized for quantum oscillators. Let's first rewrite Kuramoto in the unit vector form (see Chandra and Ott 2019 paper in Physical Review E), with $\sigma_i = (\cos \theta_i, \sin \theta_i)$ as unit vector,

$$\frac{d\sigma_i}{dt} = W_i \sigma_i + \frac{K}{N} \sum_{j=1}^N [\sigma_j - (\sigma_j \cdot \sigma_i) \sigma_i] \quad (49)$$

The first term on the rhs rotates the unit vector with angular velocity ω_i , which means that the matrix should be written as

$$W = \begin{pmatrix} 0 & \omega_i \\ -\omega_i & 0 \end{pmatrix}$$

The second term matches the original interaction term, as the lhs is

$$\dot{\theta}_i \begin{pmatrix} -\sin \theta_i \\ \cos \theta_i \end{pmatrix},$$

which is perpendicular to σ_i , the interaction term is also perpendicular to σ_i (check by taking the inproduct), so it is in the same direction. For a correspondence we only have to show that the length is $\sin(\theta_j - \theta_i)$, which is shown as follows:

$$(\sigma_j - (\sigma_j \cdot \sigma_i) \sigma_i)^2 = \sigma_j^2 + (\sigma_j \cdot \sigma_i)^2 \sigma_i^2 - 2(\sigma_j \cdot \sigma_i)^2 = 1 - (\sigma_j \cdot \sigma_i)^2 = 1 - \cos^2(\theta_j - \theta_i) = \sin^2(\theta_j - \theta_i).$$

We only need to confirm that the sign is correct.

The mean-field approximation now involves the order parameter

$$\rho = \frac{1}{N} \sum_{j=1}^N \sigma_j$$

which leads to the uncoupled (but self-consistent) equation

$$\frac{d\sigma_i}{dt} = W_i \sigma_i + K [\rho - (\rho \cdot \sigma_i) \sigma_i] \quad (50)$$

For the generalization to higher dimensions one needs to specify an ensemble of matrices W_i that are antisymmetric so that they represent (infinitesimal) rotations.

The overall goal of this project is to confirm the recently published results on synchronization as a function of coupling strength K : for odd dimensions, synchronization emerges immediately for nonzero K values whereas for even dimensions there is finite, nonzero threshold for that to occur, and, furthermore see if these results hold for small world networks (or any other networks that you may come up with).

Your analytical goals are:

- (1) Document the equivalence between the original formulation of the Kuramoto model and the generalized formulation in terms of unit vectors in 2D.
- (2) Derive and explain the outcome of the mean field approximation in the original phase formulation. You will need to assume a Lorentzian distribution of intrinsic frequencies centered around zero (see the calculation in the Pikovsky et al book).
- (3) Derive and explain the outcome of the mean field approximation for the generalized case.
- (4) Explain (supported by your numerical work specified below) the essential difference between odd and even dimensions that is responsible for the different synchronization behavior.

Your numerical goals are:

- (1) Implement the Kuramoto model (phase formulation) and validate it by comparing to the mean-field result. How many oscillators do you need to approach the analytical result (which assumes infinitely many)
- (2) Implement the generalized Kuramoto model for all dimensions, take specific care to create an appropriate ensemble for the W_i matrices
- (3) Determine the $|\rho|$ versus K curve in the range from -1 to 3.
- (4) A small world network is created by first connecting nodes in a regular network on a circle, i.e. each node is connected to the m nearest neighbors. Then for each connection there is a probability p for it to be randomly reconnected. You can vary m relative to the N and p . Are the results similar to the fully connected case you studied before? (The small world network is explained better in the Watts & Strogatz 1998 Nature paper, which includes some typical parameter setting you could use, this paper is one of the most highly cited papers in science).

1.12 Project 12: Reduction of large-scale neural network to exact mean field equations

Neural networks are comprised of a great many neurons coupled to each other by synapses, each of them represented by complicated differential equations. A fundamental question is whether we can describe a large system more or less exactly in terms of a lower dimensional system of effective variables, similar to mean-field approaches used to solve other problems in physics. In recent years such a description has emerged in non-linear dynamics that is applicable to dynamics of neural networks. The goal of this project is to get insight in the analytical correspondence between large neural network and simpler effective descriptions, implement both descriptions in terms of systems of coupled

nonlinear differential equations, validate the analytically obtained correspondence and finally study the low-dimensional description parametrically using the techniques you learned in this course. Most background information (including typical parameter settings) can be found in the paper by Montbrió, Pazo and Roxin, published in Physical Review X in 2015.

We use a quadratic integrate and fire model for the neurons in the large-scale network,

$$\dot{V}_j = V_j^2 + I_j, \quad (51)$$

together with a spiking reset, if $V_j \geq V_p$ then $V_j \leftarrow V_r$. Here V_p is the spike threshold and V_r is the reset voltage. For the analytical considerations we take the limit $V_r = -V_p \rightarrow \infty$. The driving current is composed of three terms:

$$I_j = \eta_j + Js(t) + I(t) \quad (52)$$

A stationary (constant) current offset η_j which has a Lorentzian distribution,

$$g(\eta) = \frac{1}{\pi} \frac{\Delta}{(\eta - \bar{\eta})^2 + \Delta^2},$$

with mean $\bar{\eta}$ and width Δ , a time varying current common to all neurons and a current due to synaptic inputs, scaled by a strength J which multiplies the synaptic gating variable $s(t)$, which is given by

$$s(t) = \frac{1}{N} \sum_{j=1}^N \sum_{k, \text{ with } t_j^k < t} \int_{-\infty}^t dt' a_\tau(t-t') \delta(t' - t_j^k) \quad (53)$$

here δ is the Dirac delta function and $a_\tau = \exp(-t/\tau)/\tau$ is the time course of a unitary synaptic input.

The fundamental hypothesis is that for an infinite system, the dynamics can be described by a distribution of the state variable V , as a function of time and dependent on the constant offset η specific to each neuron:

$$\rho(V|\eta, t) = \frac{1}{\pi} \frac{x(\eta, t)}{[V - y(\eta, t)]^2 + x(\eta, t)^2}. \quad (54)$$

When considering the dynamics of the mean $y(\eta, t)$ and width $x(\eta, t)$ of this distribution, a low dimensional description in terms of the mean potential v and the firing rate r can be derived:

$$\dot{r} = \frac{\Delta}{\pi} + 2rv \quad (55)$$

$$\dot{v} = v^2 + \hat{\eta} + Jr + I(t) - \pi^2 r^2 \quad (56)$$

Your analytical goals are:

- (1) Supply the details in the derivation of the reduction of the network dynamics to a low-dimensional equation (challenging).

- (2) Analytically calculate the phase diagram for the reduced system as a function of the normalized coupling $J/\sqrt{\Delta}$ and mean $\bar{\eta}/\Delta$. Determine whether the system has a single stable node, is bistable (two stable fixed points and a saddle node in between), or has a stable focus (=spiral). Draw the phase plane for an example of each one of these different system states.

Your numerical goals are:

- (1) Implement the full network.
- (2) Implement the reduced network.
- (3) Validate that the reduced and full network give similar population dynamics.
- (4) Chaos can emerge when a periodic current is injected. Use the low-dimensional system to find a chaotic regime and confirm it in the full network.